



Anhui Moon Technology Co., Ltd

Room 449, Building 3, Industrial Park, 1151 Yan'an South Road, Bengbu City, Anhui Province, China
TEL, FAX: +86-552-2079969 POST CODE: 233000

Methyl tri methyl ethyl ketoxime silane -MSDS

1. PRODUCT AND COMPANY IDENTIFICATION

1.1 Name of substance/preparation

Commercial product name: D-31(MOS/TOS=80/20)

Chemical name: Methyl tri methyl ethyl ketoxime silane

1.2 Use of substance/preparation

Industrial, crosslinker for silicone sealant

1.3 Company information

Manufacturer/distributor: Anhui Moon Technology Co., Ltd.

Room 449, Building 3, Industrial Park, 1151 Yan'an South Road, Bengbu City, Anhui Province, China

Telephone: 0086-552-2079969

Fax: 0086-552-2079969

1.4 Revision date

2019.08.01

2. HAZARDS IDENTIFICATION

2.1 Emergency overview

Combustible. May be harmful if inhaled. May be harmful if swallowed. May be harmful if absorbed through skin. Irritating to respiratory system. May cause eye and skin irritation. May cause burns. May cause allergic skin reaction. May cause irritation of the gastrointestinal tract. Will reduce the ability of the blood to transport oxygen (methemoglobinemia and anemia).

GHS Label elements, including precautionary statements



Pictogram

Signal word

Classification

Warning

Skin sensitization Category 1

2.2 Potential health effects

Ingestion: Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhoea. May cause systemic poisoning with symptoms paralleling those of inhalation.

Skin: May cause skin irritation. May be harmful if absorbed through skin. May cause allergic skin reaction. May cause systemic poisoning with symptoms paralleling those of inhalation.

Inhalation: May cause respiratory tract irritation. Vapors may be irritating to eyes, nose, throat, and lungs. The vapor may have narcotic effect. Inhalation of high vapor concentrations can cause CNS-depression and narcosis. Will reduce the ability of the blood to transport oxygen (methemoglobinemia and anemia).

Eyes: May cause eye irritation. Signs/symptoms can include redness, swelling, pain, and tearing. May cause burns.



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Medical conditions aggravated: Eye disorders, skin disorders, respiratory disorders.

2.3 Other effects

Chronic exposure: Will reduce the ability of the blood to transport oxygen (methemoglobinemia and anemia). Based on animal evidence, there is limited evidence of a carcinogenic effect. The significance of these findings for humans has not been determined.

Carcinogenicity

No components of this product present at levels greater than or equal to 0.1% is identified as a known or anticipated carcinogen by NTP, IARC, or OSHA.

3. COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Chemical characterization (substance)

CAS No.	Chemical characteristics
22984-54-9 34206-40-1	Crosslinker for silicone sealant

3.2 Hazardous ingredients

Chemical name	CAS No.	Concentration
Methyl tri methyl ethyl ketoxime silane	22984-54-9	≥78.0%
Tetrakis(methyl ethyl ketoxime)silane	34206-40-1	≥18.0%
Methyl ethyl ketoxime	96-29-7	≤1.0%
Dimer		≤1.0%

Notes: Occupational Exposure Limit(s), if available, are listed in Section 8. Harmful if swallowed.

4. FIRST AID MEASURES

4.1 General information

Take persons to a safe place. Observe self-protection for first aid.

4.2 After inhalation

Move the exposed person to fresh air at once. Keep the patient calm. Protect against loss of body heat. If breathing stops, administer artificial respiration. Use oxygen as required, provided a qualified operator is present. Seek medical advice immediately and clearly identify substance.

4.3 After contact with the skin

Remove contaminated clothes at once. Wash off with plenty of water or water and soap immediately for 10-15 minutes. In serious cases, use emergency shower immediately. Seek medical advice immediately and clearly identify substance.

4.4 After contact with the eyes

Immediately flush the contaminated eye(s) with lukewarm, gently flowing water for 15-30min while holding the eyelid(s) open. If contact lens is present, DO NOT delays irrigation or attempt to remove the lens. Do not interrupt flushing. If necessary, continue flushing during transport to emergency care facility. Take care not to rinse contaminated water into the unaffected eye or onto the face. Seek medical advice immediately. Continue to bathe eyes during transport to medical practitioner.

4.5 After swallowing



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If conscious, give several small portions of water to drink. Do not induce vomiting. Seek medical advice immediately and clearly identify substance.

4.6 Advice for the physician

Treat symptomatically.

5. FIRE FIGHTING MEASURES

5.1 Flash point

Flash point is more than 90°C (Test method: closed cup).

5.2 Suitable extinguishing media

Extinguishing powder, alcohol-resistant foam, carbon dioxide, sand.

5.3 Specific hazards during fire fighting

Combustible. Vapors may form explosive mixtures with air. Vapors are heavier than air and may spread along floors. Vapors may travel to areas away from work site before igniting/flashing back to vapor source. In case of fire hazardous decomposition products may be produced such as: carbon dioxide (CO₂), carbon monoxide (CO), oxides of nitrogen (NO_x), dense black smoke, methylethyl ketoxime (MEKO), silicone oxide, methyl ethyl ketone.

5.4 Special protective equipment for fire fighting

Wear self-contained breathing apparatus and protective suit.

5.5 Additional advice

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. Use water spray to cool unopened containers.

6. ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions

Wear personal protection equipment. Keep unprotected persons away. Ensure adequate ventilation. Remove all sources of ignition. Do not swallow. Avoid breathing vapors, mist or gas. Avoid contact with skin, eyes and clothing. If material is released indicate risk of slipping.

6.2 Environmental precautions

Prevent further leakage or spillage if safe to do so. Discharge into the environment must be avoided. Prevent product from entering drains. Do not flush into surface water or sanitary sewer system. Do not allow run-off from fire fighting to enter drains or water courses.

6.3 Methods for cleaning up

Ventilate the area. No sparking tools should be used. Use explosion-proof equipment. Do not flush away with water. For small amounts: Absorb with a liquid binding material such as diatomaceous earth and dispose of according to local/state/federal regulations. Contain larger amounts and pump up into suitable containers. Clean any slippery coating that remains using a detergent/soap solution or another biodegradable cleaner.



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7. HANDLING AND STORAGE

7.1 Handling

Precautions for safe handling

Wear personal protective equipment. Use only in well-ventilated areas. Keep container tightly closed. Protect from atmospheric moisture and water. Do not smoke. Do not swallow. Avoid breathing vapors, mist or gas. Avoid contact with skin, eyes and clothing.

Precautions against fire and explosion

Keep away from fire, sparks and heated surfaces. Keep product and empty container away from heat and sources of ignition. Take precautionary measures against static discharges. Ensure all equipment is electrically grounded before beginning transfer operations. Use explosion-proof equipment. No sparking tools should be used. No smoking.

7.2 Storage

Store in area designed for storage of flammable liquids. Protect from physical damage. Keep containers tightly closed in a dry, cool and well-ventilated place. Containers which are opened must be carefully resealed and kept upright to prevent leakage. Keep away from heat and sources of ignition. Keep away from direct sunlight. Protect from atmospheric moisture and water. Store away from incompatible substances. Container hazardous when empty. Do not pressurize, cut, weld, braze, solder, drill, grind or expose containers to heat or sources of ignition. Storage must be locked. Dispose of contents / container in accordance with local regulations

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Exposure limits

Component	CAS No.	Source	Value
Methylethyl ketoxime	96-29-7	WEEL TWA	10 ppm, 36 mg/m ³
Methylethyl ketoxime	96-29-7	Honeywell TWA	3 ppm
Methylethyl ketoxime	96-29-7	Honeywell STEL	10 ppm

8.2 Exposure in the work place limited and controlled

Protective measures: Ensure that eyewash stations and safety showers are close to the workstation location.

Engineering measures: Use with local exhaust ventilation. Prevent vapor buildup by providing adequate ventilation during and after use.

Eye protection: Do not wear contact lenses. Wear as appropriate: safety glasses with side-shields, safety goggles. If splashes are likely to occur, wear goggles or face shield, giving complete protection to eyes.

Hand protection: Solvent-resistant gloves (butyl-rubber), Neoprene gloves, Gloves must be inspected prior to use. Replace when worn.

Skin and body protection: Wear as appropriate: long sleeved clothing, gloves. If splashes are likely to occur, wear protective suit.

Respiratory protection: In case of insufficient ventilation wear suitable respiratory equipment. Use NIOSH approved respiratory protection. For rescue and maintenance work in storage tanks use



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self-contained breathing apparatus.

Hygiene measures: When using, do not eat, drink or smoke. Wash hands before breaks and immediately after handling the product. Keep working clothes separately. Remove and wash contaminated clothing before re-use. Do not swallow. Avoid breathing vapors, mist or gas. Avoid contact with skin, eyes and clothing.

8.3 Exposure to the environment limited and controlled

Prevent material from entering surface waters and soil.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state/form.....: liquid
Color.....: Colorless to lightly yellow transparent
Boiling point.....: $>300^{\circ}\text{C}$ (at 760mmHg)
Melting point.....: not determined
Flash point.....: $>90^{\circ}\text{C}$ (closed cup)
Vapor pressure.....: not determined
Density/relative density.....: $0.98\text{g/cm}^3(25^{\circ}\text{C})$
Solubility.....: hydrolyze to oxime in the presence of moisture

10. STABILITY AND REACTIVITY

10.1 General information

If stored and handled in accordance with standard industrial practices no hazardous reactions are known.

10.2 Conditions to avoid

Stable under normal conditions. Heat, flames and sparks. Keep away from direct sunlight. Protect from atmospheric moisture and water.

10.3 Materials to avoid

Acids Oxidizing agents, Metals, Iron

10.4 Hazardous thermal decomposition

Decomposes in contact with water. Possible decomposition products in case of hydrolysis are methyl ethyl ketoxime (MEKO). In case of fire hazardous decomposition products may be produced such as: carbon dioxide (CO_2), carbon monoxide (CO), oxides of nitrogen (NO_x), dense black smoke, methyl ethyl ketone, silicone oxide.

10.5 Hazardous reactions

Avoid exposure to water, strong acids and heat, especially in the presence of iron. Hazardous polymerization may occur. May react violently if in contact with electrophiles, such as ferric chloride. Stable under recommended storage conditions.



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11. TOXICOLOGICAL INFORMATION

Items	Species	Value	Time	Response	Substance
Acute oral toxicity	LD50 rat	2260-2650 mg/kg	--	--	MOS
Acute dermal toxicity	rabbit	1000-1800 mg/kg	--	--	MEKO
Acute inhalation toxicity	LC50 rat	> 4.8 mg/l	4 h	--	MEKO
Skin irritation	rabbit	--	--	slight irritation	MOS
Eye irritation	rabbit	--	--	slight irritation	MOS
Sensitisation	guinea pig	--	--	sensitization	MEKO
Repeated dose toxicity	Oral, rat,	10mg/kg/d	28 d	Blood effects, anemia	MOS
Repeated dose toxicity	rat	NOEL: 25ppm	4 wks	Blood effects, anemia	MEKO
Repeated dose toxicity	mouse	374 ppm	18months	Liver tumors	MEKO
Repeated dose toxicity	Oral rat	13 mg/kg	13 wks	--	MEKO

Additional advice: Toxicology data for the components. Based on animal evidence, there is limited evidence of a carcinogenic effect. The significance of these findings for humans has not been determined.

12. ECOLOGICAL INFORMATION

Biodegradability: Inherently biodegradable. Test substance: Methyl ethylketoxime.

Toxicity to fish: LC50, Species: *Oncorhynchus mykiss* (rainbow trout), Dose: > 120 mg/l, Exposure time: 96 h, Test substance: MOS.

Toxicity to daphnia and other aquatic invertebrates: EC50, Species: *Daphnia magna* (Water flea), Dose: > 120 mg/l, Exposure time: 48 h, Test substance: MOS.

Toxicity to algae: Biomass EC50, Species: *Pseudokirchneriella subcapitata* (green algae), Dose: 50 mg/l, Exposure time: 72 h, Test substance: MOS.

Toxicity to algae: Growth rate EC50, Species: *Pseudokirchneriella subcapitata* (green algae), Dose: 94 mg/l, Exposure time: 72 h, Test substance: MOS.

13. DISPOSAL CONSIDERATIONS

13.1 Material

Dispose of according to regulations by incineration in a special waste incinerator. Observe local/state/federal regulations.

13.2 Unclean packaging

Completely discharge containers (no tear drops, no powder rest, scraped carefully). Containers may be recycled or re-used. Observe local/state/federal regulations.



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14. TRANSPORT INFORMATION

14.1 Land transport regulations

Not dangerous goods

14.2 Transport by sea

Not dangerous goods

14.3 Transport by air

Not dangerous goods

15. REGULATORY INFORMATION

15.1 National regulations

National and local regulations must be observed. For information on labeling please refer to section 2 of this document.

15.2 Other international regulations

1907/2006 (EU): This mixture contains only ingredients which have been subject to a pre-registration according to Regulation (EC) No. 1907/2006 (REACH).

US. Toxic Substances Control Act: On TSCA Inventory

Australia. Industrial Chemical (Notification and Assessment) Act: On the inventory, or in compliance with the inventory.

Canada. Canadian Environmental Protection Act (CEPA). Domestic Substances List (DSL): All components of this product are on the Canadian DSL list.

Korea. Toxic Chemical Control Law (TCCL) List: On the inventory, or in compliance with the inventory.

Philippines. The Toxic Substances and Hazardous and Nuclear Waste Control Act: On the inventory, or in compliance with the inventory.

China. Inventory of Existing Chemical Substances: On the inventory, or in compliance with the inventory.

New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand: On the inventory, or in compliance with the inventory.

16. OTHER INFORMATION

Prepared by Anhui Moon Technology Co., Ltd..

(R) indicates Registered Trademark

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The above information relates only to the specific materials designated herein and may not be valid for such materials used in combination with any other materials or in any process or if the material is altered or processed, unless specified in the text.

-End of Material Safety Data Sheet-